Louis Berthier

PhD Candidate in Applied Mathematics & Machine Learning

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Education

École Polytechnique (CMAP) | *PhD, Applied Mathematics & Machine Learning* — Paris, France | Mar 2024 - Feb 2027
- Industrial PhD (CIFRE) with Michelin, supervised by Pr. Eric Moulines & Dr. Ahmed Shokry.
- Adaptive and conformal soft sensors for real-time quality prediction in multistage industrial processes.

Imperial College London | *MSc Advanced Computing (Machine Learning)* — London, UK | 2022 - 2023
- Merit (2:1); MSc thesis with Distinction (1st).
- Research internship at AIRL under Pr. Antoine Cully.
- Relevant courses: Machine & Deep Learning, Reinforcement Learning, Computer Vision, Natural Language Processing, Mathematics for Machine Learning, Computational Finance, Robot Learning.

IMT Mines Alès | *Diplôme d'Ingénieur (MSc), AI & Data Science* — Alès, France | 2020 - 2023
- First-class honours with jury's congratulations; ranked 2nd of 60 in the CS & AI department, top 5% of the school.
- Research internship at the CNRS under Dr. Ludovic Gardy, Pr. Christophe Hurter & Pr. Emmanuel Barbeau.
- Relevant courses: Advanced Probability & Statistics, Operations Research, Algorithms & Complexity, Machine & Deep Learning, Software Engineering.

Experience

Michelin | *ML Research Engineer (CIFRE PhD)* — Clermont-Ferrand, France | Mar 2024 - Feb 2027

Conformal and adaptive soft sensors for real-time quality prediction in industrial batch processes. Supervised by Guillaume Ramelet & Dr. Maxime Moreaud.
- Introduced Self-Organized Conformal Prediction (**NeurIPS 2026**): unsupervised group discovery with a Self-Organizing Map for group-local calibration, cutting regional coverage gaps on all 10 datasets, by 14.8% on average, for only 3% larger prediction sets.
- Co-developed Mask-Valid Conformal Prediction (with M. Hegazy, M. I. Jordan, A. Dieuleveut): binary monitoring masks that interpolate between marginal and anytime coverage, with restart validity after failures and no need for past test labels.
- Created and open-sourced TorchSOM (**JMLR 2026**), a PyTorch library for Self-Organizing Maps: 81-98% faster than MiniSom on CPU, up to about 12× faster than somoclu on GPU, and the lowest topographic error in every rectangular-map benchmark (62-100% below MiniSom).
- Developed adaptive Just-in-Time soft sensors pairing SOM similarity retrieval with online SHAP feature selection, cutting MSE by up to 61% over global baselines (**IEEE ETFA 2026**) and prediction error by over 90% under year-long drift on 35k production batches.
- Built an explainable analytics framework (frequency-ranked feature selection, SHAP interactions, multi-split conformal prediction) on 35k rubber-mixing records, cutting 316 input variables to 20 at equal accuracy (**Digit. Chem. Eng. 2026**).

Augmodo | *Computer Vision Scientist (part-time)* — Seattle, WA, USA (remote) | Aug 2025 - Present

Real-time Spatial AI for retail: open-world computer vision, with Sacha Hu, Ashley Napier, & Galen Han.
- Developed open-world VLMs that recognize products, aisles, and price tags from images alone, trained on paired images and product descriptions for identification and planogram matching.
- Used test-time augmentation to estimate the confidence of open-world classifiers (product identification, QC automation) and flag the uncertain predictions for human QC, across 100k+ images/day.
- Fine-tuned YOLO and the transformer-based RF-DETR with hard-negative mining and multi-GPU cloud training, to suppress confirmed empty-shelf false positives without degrading other shelf classes.
- Built the RGB+depth version of a classifier that checks empty-shelf alerts, with depth estimated by Depth Anything V2 as a fourth input channel, to tell true gaps from dark or shadowed shelves.
- Designed automated relabeling pipelines that update legacy bounding-box annotations at scale, adapting large datasets to newer architectures without manual re-annotation.

AIRL, Imperial College London | *ML Research Assistant* — London, UK | Apr 2023 - Sep 2023

Model-based uncertainty quantification for sample-efficient quality-diversity optimization in robotics. Supervised by Pr. Antoine Cully.
- Integrated probabilistic surrogate models into MAP-Elites to improve sample efficiency and noise robustness under costly, stochastic evaluations on a robotic-arm control task.
- Benchmarked eight surrogate and acquisition variants, and introduced two metrics for the uncertainty of quality-diversity archives.

CNRS (CerCo) | *ML Research Intern* — Toulouse, France | May 2022 - Jul 2022

Detection of pathological oscillations in epilepsy with signal processing and CNNs. Supervised by Dr. Ludovic Gardy, Pr. Christophe Hurter, & Pr. Emmanuel Barbeau.
- Trained CNN fast-ripple detectors on wavelet time-frequency images of macro- and micro-electrode EEG, improving accuracy by 5% through systematic hyperparameter tuning (**J. Neurosci. Methods 2025**).
- Applied Grad-CAM to localize high-frequency bursts in time-frequency maps, giving clinicians visual explanations anchored to known fast-ripple signatures.
- Accelerated EEG signal and image processing by 15% with Cython, and built 7 NoSQL databases for multi-channel EEG at macro and micro electrode scales for batch retrieval and parallel processing.

Skills

ML frameworks	PyTorch, JAX, TensorFlow, scikit-learn, OpenCV, PIL
Scientific stack	NumPy, pandas, Polars, SciPy, Matplotlib, Seaborn, Streamlit
Programming	Python, Cython, SQL, NoSQL, Bash, LaTeX
MLOps & HPC	Git, GitHub/GitLab CI/CD, Docker, Singularity, Kubernetes, Azure ML, SkyPilot, SLURM, Triton, Faiss
Tooling	Weights & Biases, MLflow, Pytest, Unittest, Grafana
Languages	French (native), English (fluent), Spanish (professional)

Publications

NeurIPS 2026	*Self-Organized Conformal Prediction: Reducing Regional Coverage Gaps with Unsupervised Group Discovery*. **L. Berthier**, A. Shokry, M. Moreaud, G. Ramelet, A. Dieuleveut. Accepted, to appear. [arXiv] [code]
JMLR 2026	*torchsom: The Reference PyTorch Library for Self-Organizing Maps*. **L. Berthier**, A. Shokry, M. Moreaud, G. Ramelet, E. Moulines. [paper] [code]
Digit. Chem. Eng. 2026	*Explainable Offline Process Analytics for Within-Grade Quality Variability in Industrial Rubber Mixing*. **L. Berthier**, A. Shokry, M. Moreaud, G. Ramelet, S. Desroziers, E. Moulines. Accepted, to appear. [code]
IEEE ETFA 2026	*Adaptive Soft Sensing: A Just-in-Time Approach with Self-Organizing Maps and Online Feature Selection*. **L. Berthier**, A. Shokry, M. Moreaud, G. Ramelet, E. Moulines. [paper]
ESCAPE 2025	*Knowledge Discovery in Large-Scale Batch Processes through Explainable Boosted Models and Uncertainty Quantification: Application to Rubber Mixing*. **L. Berthier**, A. Shokry, E. Moulines, G. Ramelet, S. Desroziers. [paper]
J. Neurosci. Methods 2025	*Detecting Fast-Ripples on Both Micro- and Macro-Electrodes in Epilepsy: A Wavelet-Based CNN Detector*. L. Gardy, J. Curot, L. Valton, **L. Berthier**, E. J. Barbeau, C. Hurter. [paper]
IEEE ICASSP 2023	*2DSBG: A 2D Semi Bi-Gaussian Filter Adapted for Adjacent and Multi-Scale Line Feature Detection*. B. Magnier, G. S. Shokouh, **L. Berthier**, M. Pie, A. Ruggiero. [paper]
Under review	*Where Should Locality Enter Conformal Prediction?* **L. Berthier**, E. Moulines.
Under review	*Beyond Anytime: Mask-Valid Conformal Prediction*. M. Hegazy, **L. Berthier**, M. I. Jordan, A. Dieuleveut.
In preparation	*A Unified Online Framework for Adaptive Soft Sensing in High-Dimensional Batch Processes*. **L. Berthier**, A. Shokry, M. Moreaud, G. Ramelet, E. Moulines.
Software	*torchsom: The Reference PyTorch Library for Self-Organizing Maps*. **L. Berthier**. Version 1.3.1. [code] [docs]

Presentations & Outreach

A Paper A Week (APAW) | *Research Seminar Founder & Organizer* — Michelin, Clermont-Ferrand | Apr 2024 - Jan 2026

- Founded and led a bi-monthly research seminar for 20+ researchers and data scientists, bridging ML theory and industrial applications across ML, CV, NLP, and UQ.
- Presented deep dives on *Attention Is All You Need*, *Quality-Diversity Optimization*, *HyenaDNA*, *Conformal Prediction*, and *TorchSOM*.

Conference & Invited Talks

- **IEEE ETFA** (Västerås, Sweden, 2026): *Adaptive Soft Sensing: A Just-in-Time Approach with Self-Organizing Maps and Online Feature Selection*.
- **ORAILIX team, LIX, École Polytechnique** (Palaiseau, 2026): *Wiring the Fly Brain into a World Model*, invited with team Piaget.
- **RJS 2026, 11th Young Statisticians Meeting** (Porquerolles, France, 2026): *Self-Organized Conformal Prediction*.
- **Michelin DoctoBib'Day** (Clermont-Ferrand, 2026): *A Unified Online Framework for Adaptive Soft Sensing in High-Dimensional Batch Processes*.
- **Mathematical PhD Seminar, École Polytechnique** (Palaiseau, 2025): *TorchSOM: Applications to Online Sensing and Uncertainty Quantification*.
- **ESCAPE 35** (Ghent, Belgium, 2025): *Knowledge Discovery in Large-Scale Batch Processes through Explainable Boosted Models and Uncertainty Quantification: Application to Rubber Mixing*.
- **Data Scientists Network, Michelin** (Clermont-Ferrand, 2025): *Online Sensing for Quality Monitoring*.
- **MT180, Institut Polytechnique de Paris** (Palaiseau, 2025): *Robust real-time quality monitoring and reliable adaptation of manufacturing processes in the tire industry*. Audience Award; qualified for the national round.

Posters

- **Local Dynamic Calibration via Self-Organized CP**: SIAM UQ26 (Minneapolis, 2026) and Michelin DoctoBib'Day (Clermont-Ferrand, 2026).
- **Knowledge Discovery in Large-Scale Batch Processes**: Welcome Day IP Paris (2024). Best Poster, Mathematics category.
- **A Framework for Knowledge Discovery in Rubber Mixing Processes**: Michelin Doctoral Day (Clermont-Ferrand, 2024).
- **2DSBG: A 2D Semi Bi-Gaussian Filter Adapted for Adjacent and Multi-Scale Line Feature Detection**: ICASSP 2023 (Rhodes, Greece). With B. Magnier, G. S. Shokouh, M. Pie, A. Ruggiero.

Teaching

ENSTA Paris | *Teaching Assistant* — Paris, France | Sep 2024 - Oct 2026

- Led the lab sessions of *Introduction to Machine Learning and Data Science* (20h/year for three consecutive years, 20 to 25 students per session) for first-year engineering students (B3/MSc1): data preprocessing, model training, and evaluation, under Pr. Philippe Xu.

Honours & Awards

Honours

2023 **MSc Thesis with Distinction (1st)**, Imperial College London. Invited by Pr. Antoine Cully to pursue a PhD at AIRL.

2023 **MSc with Merit (2:1)**, Imperial College London.

2023 **Diplôme d'Ingénieur with First-Class Honours (1st)**, IMT Mines Alès. Jury's congratulations; ranked 2nd of 60 in the CS & AI department, top 5% of all students.

Grants & Academic Awards

2026 **Unaite Fellowship for AI in Biology**, Paris. Selected for the inaugural cohort of this year-long international research fellowship. Project: generative & world models for EEG, with fellow researchers and Paul Hellegouarch, mentored by Dr. Quentin Garrido & Dr. Jean-Baptiste Masson. [program]

2025 **MT180 Audience Award** (€100), Institut Polytechnique de Paris; qualified for the national round.

2024 **Best PhD Poster Award** (€500), Institut Polytechnique de Paris.

2022 **International Mobility Scholarship** (€4,000), Région Occitanie.

2022 **Alumni Scholarship** (€1,300), Mines Alès Alumni.

2022 **Excellence Scholarship** (€4,000), IMT Mines Alès.

Competitions & Hackathons

2026 **2nd of 25 teams, Hack the World(s)**, Paris. Built a hierarchical JEPA world model on the complete fly-brain connectome with team *Piaget*; reposted and congratulated by Yann LeCun. [project] [report]

2026 **Selected participant, Mistral Worldwide Hackathon**, Paris. Selected from 1,700+ applicants; built *Cerno*, an AI recruitment platform with adaptive voice-AI interviews, in 36 hours. [video]

2026 **Paris Fintech Hackathon (Solve with AI)**, HEC Paris. Built *WEBI*, an AI copilot for structured bond products. [Devpost]

2024 **Winner, AI'NNOV Manufacturing IT Hackathon** (€500), Michelin.

2021 **Creativity Laureate, Communication & Media** (€80), Cora Hypermarket.

Schools & Training

2026 **MBZUAI ML Winter School**, Abu Dhabi. Representation learning and generative AI.

2025 **ECAS-SFdS School**, Fréjus. Reliable machine learning: transfer and physics-informed learning, conformal prediction.

Volunteering & Sports

EmaVisual (Photography & Video Association) | *Vice President* — Alès, France | Apr 2021 - Apr 2022

- Led a team of 10 photographers covering 25 university events; edited and produced videos and 1,000+ photos.

Bureau des Sports (Sports Association) | *Vice President* — Alès, France | Feb 2021 - Apr 2022

- Led a 25-person team running 5 tournaments across 20 sports; secured partnerships with brands including Armor Lux and Tony Parker; managed the website and IT infrastructure.

Tsiky Zanaka (Humanitarian Association) | *Tsikyst Volunteer* — Alès, France | Oct 2020 - Oct 2021

- Raised over €3,500 for solidarity projects in Madagascar through fundraising events, including cooking for 300 students.

University Handball | *Player* — France & UK | Oct 2020 - Nov 2023

- 3rd in the national university league in the UK (BUCS) and 4th in France (FFSU); won the 49th Challenge Centrale Lyon and the 19th TRAMS.